MATH 111 Calculus 1, NUU, FALL 2026

Dr. Salnikova Chekeres **Tutorial 5**

The derivative as a function and the differentiation rules (Sections 2.2–2.4, Lectures 8–9); the chain rule and implicit differentiation (Sections 2.5–2.6, Lecture 10).

Key facts to have at hand (Lectures 8–10)

- **The derivative as a function:** $f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, defined at every x for which the limit exists. The domain of f' may be *smaller* than the domain of f. Other notations: y', $\dfrac{dy}{dx}$, $\dfrac{d}{dx}f(x)$. Then $f'' = (f')'$, f''', $f^{(4)}$, …; if $s(t)$ is position, s' is velocity and s'' is acceleration.
- **Reading f' off the graph of f:** rising $\Rightarrow f' > 0$; falling $\Rightarrow f' < 0$; horizontal tangent $\Rightarrow f' = 0$; the steeper the curve, the further f' is from the axis.
- **Differentiable $\Rightarrow$ continuous** — never the other way round. Three ways to have no derivative at a: a *corner* (the one-sided slopes exist and differ), a *vertical tangent* (the difference quotient runs to $\pm\infty$), a *discontinuity*.
- **The rules:** $(c)' = 0$; $(x^n)' = nx^{n-1}$ for every real n; $(cf)' = cf'$; $(f \pm g)' = f' \pm g'$; $(fg)' = f\,g' + g\,f'$; $\left(\dfrac{f}{g}\right)' = \dfrac{g\,f' - f\,g'}{g^2}$ $(g \neq 0)$. *Rewrite before you reach for a rule:* $\sqrt{x} = x^{1/2}$, $\dfrac{5}{x^4} = 5x^{-4}$, $\dfrac{x^3+1}{x} = x^2 + \dfrac{1}{x}$.
- **Trigonometric derivatives** (radians, always): $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\tan x)' = \sec^2 x$, $(\cot x)' = -\csc^2 x$, $(\sec x)' = \sec x \tan x$, $(\csc x)' = -\csc x \cot x$. The three "co" functions carry the minus sign. Behind them: $\lim\limits_{h\to 0} \dfrac{\sin h}{h} = 1$ and $\lim\limits_{h\to 0} \dfrac{\cos h - 1}{h} = 0$.
- **The chain rule:** if $y = f(u)$ and $u = g(x)$, then $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$, i.e. $\big[f(g(x))\big]' = f'\big(g(x)\big) \cdot g'(x)$: *outside first, inside left alone, then times the inside's derivative* — and f' is fed $g(x)$, not x. For a chain of three, three factors. Special cases: $\big[g(x)^n\big]' = n\,g(x)^{n-1} g'(x)$, $\big[\sin g(x)\big]' = \cos g(x) \cdot g'(x)$.
- **Implicit differentiation:** to find y' from an equation in x and y, differentiate both sides with respect to x — *every y you differentiate brings a y'* (chain rule: $(y^3)' = 3y^2 y'$, $(\sin y)' = \cos y \cdot y'$, $(xy)' = y + xy'$) — then solve for y'. The answer may contain x and y; substitute the point *after* solving. For y'', differentiate y' again and replace y' by what you found. The method assumes the curve is a graph near the point; where it is not (the circle $x^2 + y^2 = 169$ at $(13, 0)$), no y' solves the equation.

Part I · The derivative as a function; the rules; the trigonometric derivatives (Sections 2.2–2.4) · 50 minutes

Warm-up I · True or false? (justify with one sentence or one counterexample)

1. If f is differentiable at a, then f is continuous at a.
2. If f is continuous at a, then f is differentiable at a.
3. If the graph of f has a corner at a, then f is not continuous at a.
4. Wherever the graph of f is below the x-axis, the graph of f' is below the x-axis.
5. The derivative of $\dfrac{1}{x^2}$ is $\dfrac{1}{2x}$.
6. $\dfrac{d}{dx}\big(x \cdot x^2\big) = 1 \cdot 2x = 2x$.
7. $\dfrac{d}{dx}\cos x = \sin x$.
8. The derivative of a polynomial of degree 5 is a polynomial of degree 4.
9. If $f(x) = \pi^3$, then $f'(x) = 3\pi^2$.
10. If $s(t)$ is the position of a particle at time t, then $s''(t)$ is its acceleration.

Solution.

1 True. This is the theorem of Lecture 8, with its two-line proof. **Proof.** For $x \neq a$, $f(x) - f(a) = \dfrac{f(x)-f(a)}{x-a} \cdot (x-a)$, so $\lim_{x\to a}\big[f(x)-f(a)\big] = \lim_{x\to a} \dfrac{f(x)-f(a)}{x-a} \cdot \lim_{x\to a}(x-a) = f'(a)\cdot 0 = 0$, i.e. $\lim_{x\to a} f(x) = f(a)$. □

2 False. $f(x) = |x|$ is continuous at 0, but its difference quotient at 0 is $\dfrac{|h|}{h}$, which equals 1 for $h > 0$ and -1 for $h < 0$, so $\lim_{h\to 0} \dfrac{|h|}{h}$ does not exist: no derivative. Continuity is about the values; differentiability is about the direction.

3 False. At a corner the graph is unbroken — that is what makes it a corner and not a jump. $|x|$ has a corner at 0 and is continuous there. Of the three ways to have no derivative — a corner, a vertical tangent, a discontinuity — only the third breaks the graph; the first two occur even when f is continuous.

4 False. The graph of f' records *slopes*, not heights. $f(x) = x$ is below the x-axis for $x < 0$ and rising there, so $f'(x) = 1 > 0$. Likewise $f(x) = -x^2$ at $x = 1$: the value is -1, the slope is -2 — but at $x = -1$ the value is -1 and the slope is $+2$.

5 False. Rewrite: $\dfrac{1}{x^2} = x^{-2}$, so the derivative is $-2x^{-3} = -\dfrac{2}{x^3}$. (The wrong answer comes from "dividing by the derivative of the bottom"; there is no such rule.)

6 False. $x \cdot x^2 = x^3$, whose derivative is $3x^2$, not $2x$. The derivative of a product is *not* the product of the derivatives: the product rule gives $x \cdot 2x + x^2 \cdot 1 = 3x^2$, correctly.

7 False. $(\cos x)' = -\sin x$: cosine is falling on $(0, \pi)$, where $\sin x > 0$, so its derivative must be negative there. The three "co" functions carry the minus sign.

8 True. Term by term, $(a_k x^k)' = k a_k x^{k-1}$ lowers each degree by one, and the leading term $a_5 x^5$ becomes $5a_5 x^4 \neq 0$. So the derivative has degree exactly 4.

9 False. π^3 is a number (about 31), not a power of the variable; f is constant and $f'(x) = 0$. The power rule is for x^n, not for c^n.

10 True. $s'(t) = v(t)$ is the velocity and $s''(t) = v'(t) = a(t)$ is the acceleration — the rate at which the rate is changing.

Problem 1 *The rules — and the rewriting that comes before them.*

Differentiate. Rewrite roots, reciprocals and quotients whose denominator is a single power of x before you reach for a rule, and name the rule you use.

(a) $f(x) = 2x^5 - 4x^3 + 7x - 9$

(b) $g(x) = 8\sqrt{x} - \dfrac{3}{x} + 6\sqrt[3]{x}$

(c) $F(t) = (t^2+3)(2t-5)$ (two ways: multiply out first; by the product rule)

(d) $y = \dfrac{x^3 - 4x + 6}{x^2}$ (without the quotient rule)

(e) $y = \dfrac{2x+3}{x^2+1}$

(f) $G(x) = (x-1)(x+1)(x^2+1)$

(g) For $P(x) = x^4 - 6x^2 + 2x$ find P', P'', P''', $P^{(4)}$ and $P^{(5)}$.

Solution.

(a) Sum, constant multiple, power and constant rules, term by term: $f'(x) = 2\cdot 5x^4 - 4\cdot 3x^2 + 7\cdot 1 - 0 = 10x^4 - 12x^2 + 7$.

(b) Rewrite every term as a power: $g(x) = 8x^{1/2} - 3x^{-1} + 6x^{1/3}$. Then the power rule, with $n = \frac{1}{2}$, -1, $\frac{1}{3}$:

$$g'(x) = 8\cdot\tfrac{1}{2}x^{-1/2} - 3\cdot(-1)x^{-2} + 6\cdot\tfrac{1}{3}x^{-2/3} = 4x^{-1/2} + 3x^{-2} + 2x^{-2/3} = \frac{4}{\sqrt{x}} + \frac{3}{x^2} + \frac{2}{\sqrt[3]{x^2}}.$$

(Check the sign of the middle term: for $x > 0$ the function $-3/x$ *rises* as x grows, and indeed its derivative $3/x^2$ is positive.)

(c) *First way:* multiply out, $F(t) = 2t^3 - 5t^2 + 6t - 15$, so $F'(t) = 6t^2 - 10t + 6$. *Second way:* the product rule, "the first times the derivative of the second, plus the second times the derivative of the first": $F'(t) = (t^2+3)\cdot 2 + (2t-5)\cdot 2t = 2t^2 + 6 + 4t^2 - 10t = 6t^2 - 10t + 6$. The same answer — as it must be.

(d) A quotient whose denominator is a single power of x is a sum of powers: $y = \dfrac{x^3}{x^2} - \dfrac{4x}{x^2} + \dfrac{6}{x^2} = x - 4x^{-1} + 6x^{-2}$ (for $x \neq 0$). Then $y' = 1 + 4x^{-2} - 12x^{-3} = 1 + \dfrac{4}{x^2} - \dfrac{12}{x^3}$. The quotient rule would give the same thing after much more algebra.

(e) Here the rewriting does not help, so: the quotient rule, "bottom times derivative of top, minus top times derivative of bottom, over bottom squared":

$$y' = \frac{(x^2+1)\cdot(2x+3)' - (2x+3)\cdot(x^2+1)'}{(x^2+1)^2} = \frac{(x^2+1)\cdot 2 - (2x+3)\cdot 2x}{(x^2+1)^2},$$

and the numerator is $2x^2 + 2 - 4x^2 - 6x = -2x^2 - 6x + 2$, so $y' = \dfrac{-2x^2 - 6x + 2}{(x^2+1)^2}$.

(f) $(x-1)(x+1) = x^2 - 1$ and $(x^2-1)(x^2+1) = x^4 - 1$, so $G(x) = x^4 - 1$ and $G'(x) = 4x^3$ — one line, where the product rule used twice would take ten.

(g) $P'(x) = 4x^3 - 12x + 2$; $P''(x) = 12x^2 - 12$; $P'''(x) = 24x$; $P^{(4)}(x) = 24$; $P^{(5)}(x) = 0$ — and every derivative after that is 0 too. A polynomial of degree 4 vanishes after five differentiations.

Problem 2 *Product and quotient from a table.*

Two functions f and g are differentiable, and

$$f(1) = 3, \quad f'(1) = -2, \quad g(1) = 4, \quad g'(1) = 5; \qquad f(2) = -1, \quad f'(2) = 6, \quad g(2) = 2, \quad g'(2) = -3.$$

(a) $(fg)'(1)$

(b) $\left(\dfrac{f}{g}\right)'(1)$

(c) $\left(\dfrac{g}{f}\right)'(1)$

(d) $\dfrac{d}{dx}\left[x^2 f(x)\right]$ at $x = 2$

(e) $\dfrac{d}{dx}\left[\dfrac{f(x)+g(x)}{x}\right]$ at $x = 2$

(f) The equation of the tangent line to $y = f(x)g(x)$ at $x = 2$.

Solution. No formulas for f and g are given, and none are needed: each rule uses only the values and the derivatives *at the point*.

(a) Product rule: $(fg)'(1) = f(1)\,g'(1) + g(1)\,f'(1) = 3\cdot 5 + 4\cdot(-2) = 15 - 8 = 7$.

(b) Quotient rule, bottom g: $\left(\dfrac{f}{g}\right)'(1) = \dfrac{g(1)f'(1) - f(1)g'(1)}{g(1)^2} = \dfrac{4\cdot(-2) - 3\cdot 5}{4^2} = \dfrac{-8-15}{16} = -\dfrac{23}{16}$.

(c) Now the bottom is f, so the roles swap: $\left(\dfrac{g}{f}\right)'(1) = \dfrac{f(1)g'(1) - g(1)f'(1)}{f(1)^2} = \dfrac{3\cdot 5 - 4\cdot(-2)}{3^2} = \dfrac{15+8}{9} = \dfrac{23}{9}$. Notice that this is *not* $1/(-23/16)$: the derivative of $1/u$ is not $1/u'$.

(d) Product rule with the first factor x^2: $\dfrac{d}{dx}\left[x^2 f(x)\right] = 2x\,f(x) + x^2 f'(x)$, and at $x = 2$: $2\cdot 2\cdot(-1) + 2^2\cdot 6 = -4 + 24 = 20$.

(e) Let $u(x) = f(x) + g(x)$; then $u(2) = -1 + 2 = 1$ and $u'(2) = f'(2) + g'(2) = 6 - 3 = 3$. Quotient rule with bottom x: $\left(\dfrac{u}{x}\right)' = \dfrac{x\,u'(x) - u(x)\cdot 1}{x^2}$, and at $x = 2$: $\dfrac{2\cdot 3 - 1}{4} = \dfrac{5}{4}$. (Or: $u/x = u\cdot x^{-1}$ and the product rule — same number.)

(f) The point: $y(2) = f(2)g(2) = (-1)\cdot 2 = -2$. The slope: $(fg)'(2) = f(2)g'(2) + g(2)f'(2) = (-1)(-3) + 2\cdot 6 = 3 + 12 = 15$. The tangent: $y - (-2) = 15(x-2)$, that is $y = 15x - 32$.

Problem 3 *Trigonometric derivatives.*

(a) $y = x^3 \cos x$

(b) $y = \dfrac{\sin x + \cos x}{\sin x - \cos x}$ (simplify the answer)

(c) $y = 4\tan x - 3\sec x$

(d) $y = x\cos x - \sin x$ (simplify the answer)

(e) Show that $y = 4\sin x + 5\cos x$ satisfies $y'' + y = 0$.

(f) Find all points of the graph of $y = \sin x + \sqrt{3}\cos x$, $0 \le x \le 2\pi$, at which the tangent line is horizontal.

(g) (Two of the four formulas that Lecture 9 left as a homework exercise.) Starting from $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$, derive the formulas for $(\sec x)'$ and $(\cot x)'$ by the quotient rule.

Solution.

(a) Product rule: $y' = 3x^2 \cdot \cos x + x^3 \cdot (-\sin x) = 3x^2\cos x - x^3 \sin x$.

(b) Quotient rule, with $(\sin x + \cos x)' = \cos x - \sin x$ and $(\sin x - \cos x)' = \cos x + \sin x$:

$$y' = \frac{(\sin x - \cos x)(\cos x - \sin x) - (\sin x + \cos x)(\cos x + \sin x)}{(\sin x - \cos x)^2}$$

$$= \frac{-(\sin x - \cos x)^2 - (\sin x + \cos x)^2}{(\sin x - \cos x)^2}.$$

Expand the numerator: $-(\sin^2 x - 2\sin x\cos x + \cos^2 x) - (\sin^2 x + 2\sin x\cos x + \cos^2 x) = -1 - 1 = -2$, using $\sin^2 x + \cos^2 x = 1$ twice (the cross terms cancel). So $y' = -\dfrac{2}{(\sin x - \cos x)^2}$ — defined exactly where y is, $\sin x \ne \cos x$. (Since $(\sin x - \cos x)^2 = 1 - \sin 2x$, one can also write $y' = -\dfrac{2}{1 - \sin 2x}$.)

(c) Term by term, from the table: $y' = 4\sec^2 x - 3\sec x \tan x$.

(d) Product rule on $x\cos x$: $(x\cos x)' = 1 \cdot \cos x + x \cdot (-\sin x) = \cos x - x\sin x$. Then $y' = \cos x - x\sin x - \cos x = -x\sin x$. The two $\cos x$ cancel.

(e) $y' = 4\cos x - 5\sin x$ and $y'' = -4\sin x - 5\cos x = -(4\sin x + 5\cos x) = -y$. So $y'' + y = 0$. $\square$ (Every $a\sin x + b\cos x$ does this: differentiating twice brings each of sin and cos back with a minus sign.)

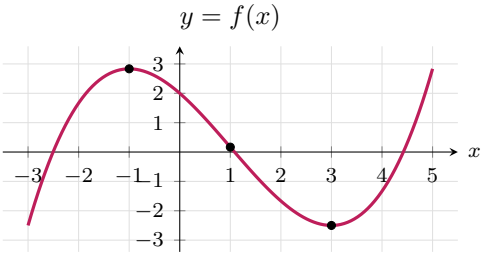
(f) A horizontal tangent means $y' = 0$: $y' = \cos x - \sqrt{3}\sin x = 0$, i.e. $\cos x = \sqrt{3}\sin x$, i.e. $\tan x = \dfrac{1}{\sqrt{3}}$ ($\cos x = 0$ is not a solution, since then $\sin x = \pm 1 \ne 0$). In $[0, 2\pi]$: $x = \dfrac{\pi}{6}$ and $x = \dfrac{\pi}{6} + \pi = \dfrac{7\pi}{6}$. The points: $y\left(\frac{\pi}{6}\right) = \frac{1}{2} + \sqrt{3} \cdot \frac{\sqrt{3}}{2} = 2$ and $y\left(\frac{7\pi}{6}\right) = -\frac{1}{2} - \frac{3}{2} = -2$: the points $\left(\frac{\pi}{6}, 2\right)$ and $\left(\frac{7\pi}{6}, -2\right)$, the top and the bottom of the wave.

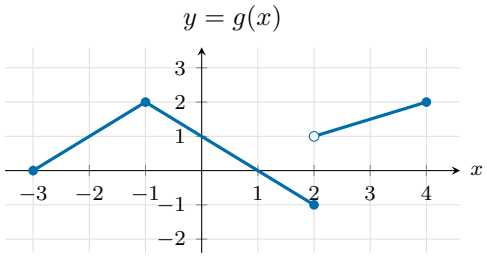
(g) $\sec x = \dfrac{1}{\cos x}$, quotient rule with top 1 and bottom $\cos x$: $(\sec x)' = \dfrac{\cos x \cdot 0 - 1 \cdot (-\sin x)}{\cos^2 x} = \dfrac{\sin x}{\cos^2 x} = \dfrac{1}{\cos x} \cdot \dfrac{\sin x}{\cos x} = \sec x \tan x$.

$\cot x = \dfrac{\cos x}{\sin x}$: $(\cot x)' = \dfrac{\sin x \cdot (-\sin x) - \cos x \cdot \cos x}{\sin^2 x} = \dfrac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = -\dfrac{1}{\sin^2 x} = -\csc^2 x$. (tan and csc are derived in the same way; csc gets the minus sign of a "co" function.)

Problem 4 *Reading f' off the graph of f; where there is no derivative.*

The graphs of two functions f and g are shown: f is defined for every x (only the part over $[-3,5]$ is drawn), and g is defined on $[-3,4]$ only.





(a) For f: on which intervals is $f' > 0$, on which $f' < 0$, and where is $f' = 0$? At which x is f' smallest (the descent steepest)? Sketch the graph of f'.

(b) For g ($g(2) = -1$, and the open dot is not on the graph): sketch the graph of g', with its domain.

(c) At which numbers in $(-3,4)$ is g not differentiable? Give the reason for each, and for the corner compute the two one-sided limits of the difference quotient.

(d) Show from the definition that $k(x) = \sqrt[3]{x-1}$ has a vertical tangent at $x = 1$: compute the limit $\lim_{h\to 0} \dfrac{k(1+h)-k(1)}{h}$ and say what it tells you.

Solution.

(a) Read slopes, not heights. Left of $x = -1$ the curve rises: $f' > 0$ on $(-\infty, -1)$. Between -1 and 3 it falls: $f' < 0$ on $(-1, 3)$. Right of 3 it rises again: $f' > 0$ on $(3, \infty)$. At the top $(-1, 2.8)$ and the bottom $(3, -2.5)$ the tangent is horizontal: $f'(-1) = f'(3) = 0$. The descent is steepest midway between the top and the bottom, at $x = 1$ (the marked point), where the tangent has slope -2. So the graph of f' is a U-shaped curve through $(-1, 0)$ and $(3, 0)$ with its lowest point at $(1, -2)$, positive outside $[-1, 3]$ and climbing steeply at both ends ($f'(-3) = f'(5) = 6$). (For the record, $f(x) = \frac{1}{6}x^3 - \frac{1}{2}x^2 - \frac{3}{2}x + 2$ and $f'(x) = \frac{1}{2}x^2 - x - \frac{3}{2} = \frac{1}{2}(x+1)(x-3)$.)

(b) g is made of three line segments, and the derivative of a line segment is its slope: on $(-3, -1)$ the slope is 1; on $(-1, 2)$ it is -1; on $(2, 4)$ it is $\frac{1}{2}$. So the graph of g' consists of three horizontal pieces, at heights 1, -1 and $\frac{1}{2}$, with open dots at $x = -1$ and $x = 2$: those two numbers are *not* in the domain of g' (part (c)), and neither are the endpoints -3 and 4, where g is defined on one side only. The domain of g' is $(-3, -1) \cup (-1, 2) \cup (2, 4)$.

(c) At $x = -1$ the graph has a corner. The two one-sided limits of the difference quotient, with $g(-1) = 2$: from the left $g(-1+h) = (-1+h) + 3 = 2 + h$, and from the right $g(-1+h) = 1 - (-1+h) = 2 - h$, so

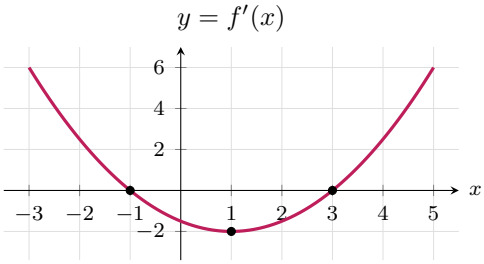
$$\lim_{h\to 0^-} \frac{g(-1+h)-g(-1)}{h} = \lim_{h\to 0^-} \frac{h}{h} = \lim_{h\to 0^-} 1 = 1, \qquad \lim_{h\to 0^+} \frac{g(-1+h)-g(-1)}{h} = \lim_{h\to 0^+} \frac{-h}{h} = -1.$$

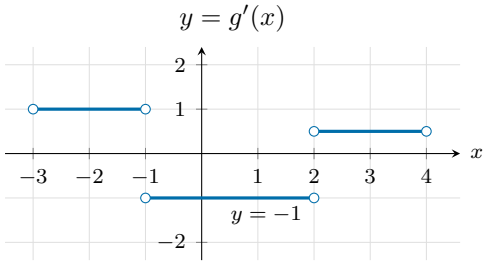
The one-sided limits differ ($1 \neq -1$), so the two-sided limit does not exist: $g'(-1)$ does not exist, although g is continuous there. At $x = 2$ the graph jumps: $g(2) = -1$ but $g(x) \to 1$ as $x \to 2^+$, so g is not continuous at 2 — and "differentiable $\Rightarrow$ continuous" says that a function which is not continuous cannot be differentiable. Two numbers, two different reasons.

(d) $k(1) = 0$ and $k(1+h) = \sqrt[3]{h}$, so

$$\lim_{h\to 0} \frac{k(1+h)-k(1)}{h} = \lim_{h\to 0} \frac{\sqrt[3]{h}}{h} = \lim_{h\to 0} \frac{h^{1/3}}{h} = \lim_{h\to 0} h^{-2/3} = \lim_{h\to 0} \frac{1}{\sqrt[3]{h^2}} = +\infty,$$

because $\sqrt[3]{h^2}$ is positive for every $h \neq 0$ (from either side, since $h^2 > 0$) and tends to 0. The limit is not a number, so $k'(1)$ does not exist; the secant lines become steeper and steeper and the tangent line at $(1, 0)$ is the *vertical* line $x = 1$ — the second of the three ways to have no derivative. □





Left: f' is zero under the points where f has a horizontal tangent, and lowest where f falls fastest. Right: g' is the slope of each piece, and has no value at the corner, at the jump or at the two endpoints.

Problem 5 *Differentiable, or only continuous?*

(a) Find the constants a and b for which $f(x) = \begin{cases} x^2 - 2x, & x \le 3, \\ ax + b, & x > 3 \end{cases}$ is differentiable at every number.

(b) For which c is $g(x) = \begin{cases} cx^2, & x \le 1, \\ 3x - 1, & x > 1 \end{cases}$ continuous at 1? For that c, is g differentiable at 1? (Compare the one-sided slopes.)

(c) Show from the definition that $p(x) = x|x|$ is differentiable at 0, and find $p'(0)$. (Then write a formula for $p'(x)$ for every x.)

(d) At which numbers is $q(x) = |x^2 - 4x|$ not differentiable? Give the one-sided slopes there.

Solution.

(a) Away from 3 each piece is a polynomial, so only $x = 3$ needs checking, and two things must hold there. *Continuity* (differentiable $\Rightarrow$ continuous, so this comes first): the left piece gives $f(3) = 9 - 6 = 3$, the right piece tends to $3a + b$ as $x \to 3^+$, so $3a + b = 3$. *The same slope from both sides:* the left piece has derivative $2x - 2$, which is 4 at $x = 3$; the right piece has derivative a. So $a = 4$, and then $b = 3 - 3a = 3 - 12 = -9$. Check: with $f(x) = 4x - 9$ for $x > 3$, the right-hand limit at 3 is $12 - 9 = 3$ and the slope is 4. The two pieces meet without a corner — the line $y = 4x - 9$ is the tangent to the parabola at $(3, 3)$.

(b) Continuity at 1: $g(1) = c$ from the left piece, and $3x - 1 \to 2$ as $x \to 1^+$, so $c = 2$. With $c = 2$: the slope from the left is $(2x^2)' = 4x = 4$ at $x = 1$; from the right it is $(3x - 1)' = 3$. Different one-sided slopes — g is continuous at 1 but *not* differentiable there: a corner. (Continuity can be arranged with one constant; differentiability needs a second one, as in (a).)

(c) $p(0) = 0$, so $p'(0) = \lim_{h \to 0} \dfrac{p(0 + h) - p(0)}{h} = \lim_{h \to 0} \dfrac{h\,|h|}{h} = \lim_{h \to 0} |h| = 0$ (the h cancels because $h \neq 0$ inside the limit). So $p'(0) = 0$: differentiable at 0, with a horizontal tangent. Elsewhere: for $x > 0$, $p(x) = x \cdot x = x^2$ and $p'(x) = 2x$; for $x < 0$, $p(x) = x \cdot (-x) = -x^2$ and $p'(x) = -2x$. All three cases in one formula: $p'(x) = 2|x|$. $\square$ (So p' itself has a corner at 0 — Challenge 1.)

(d) $x^2 - 4x = x(x - 4)$ is negative exactly on $(0, 4)$, so $q(x) = x^2 - 4x$ for $x \le 0$ or $x \ge 4$, and $q(x) = -(x^2 - 4x) = 4x - x^2$ for $0 < x < 4$. Away from 0 and 4 each piece is a polynomial, differentiable, with $q'(x) = 2x - 4$ outside $[0, 4]$ and $q'(x) = 4 - 2x$ inside. At $x = 0$: from the left (outside) the slope is $2 \cdot 0 - 4 = -4$, from the right (inside) it is $4 - 0 = 4$. At $x = 4$: from the left (inside) $4 - 8 = -4$, from the right (outside) $8 - 4 = 4$. At both numbers the one-sided slopes differ, so q is not differentiable there: two corners, where the part of the graph of $x^2 - 4x$ below the x-axis was reflected above it.

Problem 6 *The derivative as a function, from the definition.*

Find $f'(x)$ from the definition, $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$, keeping "lim" in front of every line until the limit is taken. State the domain of f and the domain of f'.

(a) $f(x) = 3x^2 - x + 4$

(b) $f(x) = \dfrac{1}{x+3}$

(c) $f(x) = \sqrt{x-2}$

(d) $f(x) = x^3 - 4x$; then find $f''(x)$ the same way.

Solution.

(a) $f(x+h) = 3(x+h)^2 - (x+h) + 4 = 3x^2 + 6xh + 3h^2 - x - h + 4$, so $f(x+h) - f(x) = 6xh + 3h^2 - h$, and

$$f'(x) = \lim_{h\to 0} \frac{6xh + 3h^2 - h}{h} = \lim_{h\to 0} (6x + 3h - 1) = 6x - 1.$$

f and f' are both defined for every real x. (The power rule confirms it: $6x - 1$.)

(b) Over a common denominator:

$$f'(x) = \lim_{h\to 0} \frac{1}{h}\Big(\frac{1}{x+h+3} - \frac{1}{x+3}\Big) = \lim_{h\to 0} \frac{1}{h} \cdot \frac{(x+3) - (x+h+3)}{(x+h+3)(x+3)}$$

$$= \lim_{h\to 0} \frac{-h}{h\,(x+h+3)(x+3)} = \lim_{h\to 0} \frac{-1}{(x+h+3)(x+3)} = -\frac{1}{(x+3)^2}.$$

Domain of f: all $x \neq -3$; domain of f': the same. (Check: $(x+3)^{-1}$ has derivative $-(x+3)^{-2}$.)

(c) A root: multiply by the conjugate.

$$f'(x) = \lim_{h\to 0} \frac{\sqrt{x+h-2} - \sqrt{x-2}}{h} = \lim_{h\to 0} \frac{(x+h-2) - (x-2)}{h\big(\sqrt{x+h-2} + \sqrt{x-2}\big)}$$

$$= \lim_{h\to 0} \frac{h}{h\big(\sqrt{x+h-2} + \sqrt{x-2}\big)} = \lim_{h\to 0} \frac{1}{\sqrt{x+h-2} + \sqrt{x-2}} = \frac{1}{2\sqrt{x-2}},$$

valid when $x - 2 > 0$. The domain of f is $[2, \infty)$ (the root needs $x - 2 \geq 0$); the domain of f' is $(2, \infty)$: at $x = 2$, $\lim_{h\to 0^+} \dfrac{\sqrt{h} - 0}{h} = \lim_{h\to 0^+} \dfrac{1}{\sqrt{h}} = +\infty$ (and $h < 0$ is not allowed, since f is not defined left of 2), so $f'(2)$ does not exist — the graph starts at $(2, 0)$ with a vertical tangent. The domain of f' is *smaller* than the domain of f.

(d) $(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3$, so $f(x+h) - f(x) = 3x^2h + 3xh^2 + h^3 - 4h$ and

$$f'(x) = \lim_{h\to 0} \frac{3x^2h + 3xh^2 + h^3 - 4h}{h} = \lim_{h\to 0} (3x^2 + 3xh + h^2 - 4) = 3x^2 - 4.$$

Now the same recipe on f': $f'(x+h) - f'(x) = 3(x+h)^2 - 3x^2 = 6xh + 3h^2$, so

$$f''(x) = \lim_{h\to 0} \frac{6xh + 3h^2}{h} = \lim_{h\to 0} (6x + 3h) = 6x.$$

All domains are $\mathbb{R}$.

Part II · The chain rule and implicit differentiation (Sections 2.5–2.6) · 50 minutes

Warm-up II · True or false? (justify with one sentence or one counterexample)

1. $\dfrac{d}{dx}\sin(x^2) = \cos(x^2)$.
2. $\dfrac{d}{dx}\sin^2 x = 2\sin x\cos x$.
3. $\dfrac{d}{dx}(x^2+1)^{10} = 10(x^2+1)^9$.
4. $\dfrac{d}{dx}\sqrt{x^2+1} = \dfrac{x}{\sqrt{x^2+1}}$.
5. If $f'(2)=3$, $g(1)=2$ and $g'(1)=5$, then $(f\circ g)'(1)=15$.
6. $(f\circ g)'(x) = f'(x)\,g'(x)$ for all differentiable f and g.
7. On the circle $x^2+y^2=25$ the tangent line at $(3,4)$ has slope $-\dfrac{3}{4}$.
8. When y is a function of x, $\dfrac{d}{dx}\big(y^3\big) = 3y^2$.
9. When y is a function of x, $\dfrac{d}{dx}\big(\sin y\big) = \cos y\cdot y'$.
10. When y is a function of x, $\dfrac{d}{dx}\big(xy\big) = x\,y'$.

Solution.

1 False. $\sin(x^2)$ is a composition: outside sin, inside x^2. The chain rule gives $\cos(x^2)\cdot 2x$. Check at a point, as Lecture 10 did for $\sin(x^3)$: at $x=0$ the true slope is $\lim_{h\to0}\dfrac{\sin(h^2)-0}{h} = \lim_{h\to0}\dfrac{\sin(h^2)}{h^2}\cdot h = 1\cdot 0 = 0$, but $\cos(0^2)=1$. Something was missing — the inside's derivative.

2 True. $\sin^2 x = (\sin x)^2$: outside the square, inside $\sin x$; so $2\sin x\cdot(\sin x)' = 2\sin x\cos x$ $(=\sin 2x)$.

3 False. The inside is x^2+1, with derivative $2x$: $\dfrac{d}{dx}(x^2+1)^{10} = 10(x^2+1)^9\cdot 2x = 20x(x^2+1)^9$. The power rule alone is for x^n, where the inside is x and its derivative is 1 — which is why nothing seems to be missing in that case.

4 True. $\sqrt{x^2+1} = (x^2+1)^{1/2}$; derivative $\frac{1}{2}(x^2+1)^{-1/2}\cdot 2x = \dfrac{x}{\sqrt{x^2+1}}$.

5 True. $(f\circ g)'(1) = f'\big(g(1)\big)\cdot g'(1) = f'(2)\cdot 5 = 3\cdot 5 = 15$. The outside derivative is taken at $g(1)=2$, and that value is exactly the one given.

6 False. The correct rule is $(f\circ g)'(x) = f'\big(g(x)\big)\,g'(x)$: f' is fed $g(x)$, not x. Example: $f(x)=x^2$, $g(x)=x+1$. Then $f(g(x)) = (x+1)^2$ has derivative $2(x+1)$, but $f'(x)g'(x) = 2x\cdot 1 = 2x$. They differ at every x.

7 True. Differentiate $x^2+y^2=25$: $2x+2y\,y'=0$, so $y' = -\dfrac{x}{y}$, and at $(3,4)$ this is $-\dfrac{3}{4}$. (The radius to $(3,4)$ has slope $\frac{4}{3}$, and the tangent is perpendicular to it: $-\frac{3}{4}$. Both ways agree.)

8 False. y depends on x, so y^3 is a composition: outside the cube, inside y. The chain rule gives $3y^2\cdot y'$. Forgetting the y' is the implicit-differentiation version of item 1.

9 True. Outside sin, inside y: $\cos y\cdot y'$.

10 False. xy is a product of two functions of x, so the product rule applies: $(xy)' = 1\cdot y + x\cdot y' = y + x\,y'$. (Only the y that is differentiated brings a y'; the x is differentiated to 1.)

Problem 7 *The chain rule.*

Differentiate. For each function say what the *outside* is and what the *inside* is before you write anything.

(a) $y = (3x^2 - 5)^4$

(b) $y = \sqrt{5x^2 + 1}$

(c) $y = \dfrac{1}{(x^2 + 2x)^3}$

(d) $y = \tan(x^3)$

(e) $y = x^2(1 - 2x)^5$ (simplify: take out the common factor)

(f) $y = \left(\dfrac{x - 1}{x + 2}\right)^3$

Solution. In every part: name the outside function and the inside function, differentiate the outside *with the inside left alone*, then multiply by the derivative of the inside.

(a) Outside u^4, inside $u = 3x^2 - 5$. $y' = 4(3x^2 - 5)^3 \cdot (3x^2 - 5)' = 4(3x^2 - 5)^3 \cdot 6x = 24x(3x^2 - 5)^3$.

(b) Outside $\sqrt{u} = u^{1/2}$, inside $u = 5x^2 + 1$. $y' = \dfrac{1}{2}(5x^2 + 1)^{-1/2} \cdot 10x = \dfrac{10x}{2\sqrt{5x^2 + 1}} = \dfrac{5x}{\sqrt{5x^2 + 1}}$.

(c) Rewrite first: $y = (x^2 + 2x)^{-3}$. Outside u^{-3}, inside $u = x^2 + 2x$. $y' = -3(x^2 + 2x)^{-4} \cdot (2x + 2) = -\dfrac{3 \cdot 2(x + 1)}{(x^2 + 2x)^4} = -\dfrac{6(x + 1)}{(x^2 + 2x)^4}$. (No quotient rule was needed: a 1 on top is a negative power.)

(d) Outside $\tan u$, inside $u = x^3$. $y' = \sec^2(x^3) \cdot 3x^2 = 3x^2 \sec^2(x^3)$. The inside stays inside: it is $\sec^2(x^3)$, not $\sec^2 x$.

(e) A product, whose second factor is a chain. Product rule first: $y' = (x^2)'(1-2x)^5 + x^2\left[(1-2x)^5\right]' = 2x(1 - 2x)^5 + x^2 \cdot 5(1 - 2x)^4 \cdot (-2)$, the last factor -2 being the derivative of the inside $1 - 2x$. Now take out the common factor $2x(1 - 2x)^4$:

$$y' = 2x(1 - 2x)^4\big[(1 - 2x) - 5x\big] = 2x(1 - 2x)^4(1 - 7x).$$

(The zeros of y' are now visible: $x = 0$, $x = \frac{1}{7}$, $x = \frac{1}{2}$ — the horizontal tangents.)

(f) Outside u^3, inside the quotient $u = \dfrac{x - 1}{x + 2}$. By the quotient rule, $u' = \dfrac{(x + 2) \cdot 1 - (x - 1) \cdot 1}{(x + 2)^2} = \dfrac{3}{(x + 2)^2}$. So

$$y' = 3\Big(\frac{x - 1}{x + 2}\Big)^2 \cdot \frac{3}{(x + 2)^2} = \frac{9(x - 1)^2}{(x + 2)^4}.$$

Problem 8 *Chains of three.*

Peel from the outside in; every layer contributes one factor. (Parts (a) and (b) are the two that Lecture 10's chain-of-three slide set for the tutorial.)

(a) $y = \sqrt{1 + \sin 3x}$

(b) $y = \cos^2(5x)$ (simplify with a double-angle formula)

(c) $y = (1 + \tan 2x)^3$

(d) $y = \cos\Big(\sqrt{1 + x^2}\Big)$

Solution.

(a) Three layers: outside $\sqrt{u}$, middle $1 + \sin v$, inside $v = 3x$. Peel: $\dfrac{1}{2\sqrt{1 + \sin 3x}}$ (derivative of the outside, evaluated at the rest) $\times$ $\cos 3x$ (derivative of the middle, at the inside) $\times$ 3 (derivative of the inside). So $y' = \dfrac{3\cos 3x}{2\sqrt{1 + \sin 3x}}$.

(b) $\cos^2(5x) = (\cos 5x)^2$: outside u^2, middle $\cos v$, inside $v = 5x$. $y' = 2\cos 5x \cdot (-\sin 5x) \cdot 5 = -10 \sin 5x \cos 5x$, and since $2 \sin\theta \cos\theta = \sin 2\theta$, this is $-5 \sin 10x$.

(c) Outside u^3, middle $1+\tan v$, inside $v = 2x$: $y' = 3(1+\tan 2x)^2 \cdot \sec^2(2x) \cdot 2 = 6\sec^2(2x)(1+\tan 2x)^2$.

(d) Outside $\cos u$, middle $\sqrt{v}$, inside $v = 1+x^2$: $y' = -\sin\big(\sqrt{1+x^2}\big)\cdot\dfrac{1}{2\sqrt{1+x^2}}\cdot 2x = -\dfrac{x\sin\sqrt{1+x^2}}{\sqrt{1+x^2}}$.

In Leibniz's notation, with $y = \cos u$, $u = \sqrt{v}$, $v = 1+x^2$: $\dfrac{dy}{dx} = \dfrac{dy}{du}\dfrac{du}{dv}\dfrac{dv}{dx}$ — three factors for three links.

Problem 9 *The chain rule from a table.*

f and g are differentiable, and

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	−2	2	4
2	1	5	3	−1
3	2	6	1	3

Find: (a) $(f\circ g)'(1)$; (b) $(g\circ f)'(3)$; (c) $\dfrac{d}{dx}\big[f(x)\big]^3$ at $x=1$; (d) $\dfrac{d}{dx}f(2x-1)$ at $x=2$; (e) $\dfrac{d}{dx}\sqrt{g(x)}$ at $x=3$; (f) $(f\circ f)'(2)$; (g) $(g\circ g\circ g)'(1)$.

Solution. The rule is $(f\circ g)'(a) = f'\big(g(a)\big)\cdot g'(a)$: the inside derivative at a, the outside derivative at $g(a)$ — f' *wants* $g(a)$, *not* a. Read the table twice: once for the inside, once for the point at which to take the outside.

(a) $g(1) = 2$, so $(f\circ g)'(1) = f'(2)\cdot g'(1) = 5\cdot 4 = 20$. (The wrong point, $f'(1)g'(1) = -8$, is the trap.)

(b) Now g is outside and f inside: $f(3) = 2$, so $(g\circ f)'(3) = g'(2)\cdot f'(3) = (-1)\cdot 6 = -6$.

(c) Outside u^3, inside f: $3f(x)^2 f'(x)$, and at $x=1$: $3\cdot 3^2\cdot(-2) = -54$.

(d) Outside f, inside $2x-1$: $f'(2x-1)\cdot 2$. At $x=2$ the inside is $2\cdot 2-1 = 3$, so the value is $f'(3)\cdot 2 = 6\cdot 2 = 12$.

(e) Outside $\sqrt{u}$, inside g: $\dfrac{g'(x)}{2\sqrt{g(x)}}$, and at $x=3$: $\dfrac{3}{2\sqrt{1}} = \dfrac{3}{2}$.

(f) $f(2) = 1$, so $(f\circ f)'(2) = f'(1)\cdot f'(2) = (-2)\cdot 5 = -10$.

(g) Three links: $(g\circ g\circ g)'(1) = g'\big(g(g(1))\big)\cdot g'\big(g(1)\big)\cdot g'(1)$. With $g(1)=2$ and $g(2)=3$: $g'(3)\cdot g'(2)\cdot g'(1) = 3\cdot(-1)\cdot 4 = -12$.

Problem 10 *Implicit differentiation.*

First check that the point is on the curve; then differentiate — every y you differentiate brings a y' — and solve for y'; only then substitute the point.

(a) $3x^2 + y^2 = 7$: find y' at $(1,2)$.

(b) $x^2 + xy + 2y^2 = 28$: find y' in terms of x and y, and the equation of the tangent line at $(2,3)$.

(c) On the curve of (b), find the points at which the tangent line is horizontal.

(d) $x^2y + y^3 = 10$: find y' at $(1,2)$.

(e) $y + \sin y = x$: find y' in terms of y, and its value at $(0,0)$.

Solution.

(a) The point is on the curve: $3\cdot 1 + 4 = 7$. Differentiate both sides with respect to x: $6x + 2y\,y' = 0$ (the y^2 is a chain, outside the square, inside y). So $y' = -\dfrac{6x}{2y} = -\dfrac{3x}{y}$, and at $(1,2)$: $y' = -\dfrac{3}{2}$.

(b) On the curve: $4+6+18 = 28$. Differentiate term by term: $(x^2)' = 2x$; $(xy)' = y + x\,y'$ (product rule); $(2y^2)' = 4y\,y'$; $(28)' = 0$. So

$$2x + y + x\,y' + 4y\,y' = 0 \quad\Longrightarrow\quad y'(x+4y) = -(2x+y) \quad\Longrightarrow\quad y' = -\frac{2x+y}{x+4y}.$$

At $(2,3)$: $y' = -\dfrac{4+3}{2+12} = -\dfrac{7}{14} = -\dfrac{1}{2}$. The tangent: $y - 3 = -\dfrac{1}{2}(x-2)$, that is $y = -\dfrac{1}{2}x + 4$.

(c) A horizontal tangent means $y' = 0$, i.e. the numerator $2x + y = 0$ (while the denominator $x + 4y \neq 0$). So $y = -2x$. Such a point must also lie on the curve: $x^2 + x(-2x) + 2(-2x)^2 = x^2 - 2x^2 + 8x^2 = 7x^2 = 28$, so $x^2 = 4$, $x = \pm 2$, and $y = -2x = \mp 4$. The points are $(2,-4)$ and $(-2,4)$; at both, $x + 4y = 2 - 16 = -14 \neq 0$ and $-2 + 16 = 14 \neq 0$, so y' is indeed 0 there. (The curve is an ellipse, tilted; these are its top and bottom.)

(d) On the curve: $1 \cdot 2 + 8 = 10$. Differentiate: $(x^2y)' = 2xy + x^2y'$ (product rule, and the y brings a y'); $(y^3)' = 3y^2y'$. So $2xy + x^2y' + 3y^2y' = 0$, $y'(x^2 + 3y^2) = -2xy$, $y' = -\dfrac{2xy}{x^2 + 3y^2}$. At $(1,2)$: $y' = -\dfrac{4}{1+12} = -\dfrac{4}{13}$.

(e) Differentiate $y + \sin y = x$: $y' + \cos y \cdot y' = 1$ — the $\sin y$ is a chain, outside sin, inside y. So $y'(1 + \cos y) = 1$ and $y' = \dfrac{1}{1+\cos y}$. At $(0,0)$ (which is on the curve: $0+0=0$): $y' = \dfrac{1}{1+1} = \dfrac{1}{2}$. (This curve cannot be solved for y by any formula — y *is* a function of x here, but there is no expression for it — and the method does not need one.)

Problem 11 *The folium, continued.*

Lecture 10 differentiated the folium of Descartes, $x^3 + y^3 = 9xy$, and found $y' = \dfrac{3y - x^2}{y^2 - 3x}$.

(a) Find the point of the folium in the first quadrant, other than the origin, at which the tangent line is horizontal. (Lecture 10 left this to the tutorial: $y' = 0$ means $3y = x^2$; substitute into the equation of the curve.)

(b) Where is the tangent vertical? (Use the symmetry of the curve in the line $y = x$, or do the computation again with $y^2 = 3x$.)

(c) Find the slope and the equation of the tangent line at the point $\left(\frac{9}{2}, \frac{9}{2}\right)$, which lies on the curve. What is special about the direction?

Solution.

(a) Horizontal tangent: $y' = 0$, so the numerator $3y - x^2 = 0$, i.e. $y = \dfrac{x^2}{3}$. The point must also be on the curve: substitute,

$$x^3 + \left(\frac{x^2}{3}\right)^3 = 9x \cdot \frac{x^2}{3} \quad \Longrightarrow \quad x^3 + \frac{x^6}{27} = 3x^3 \quad \Longrightarrow \quad \frac{x^6}{27} = 2x^3.$$

Away from the origin $x \neq 0$, so divide by x^3: $x^3 = 54$, $x = \sqrt[3]{54} = \sqrt[3]{27 \cdot 2} = 3\sqrt[3]{2} \approx 3.78$. Then $y = \dfrac{x^2}{3} = \dfrac{9\sqrt[3]{4}}{3} = 3\sqrt[3]{4} \approx 4.76$. Check that the denominator of y' is not zero there: $y^2 - 3x = 9\sqrt[3]{16} - 9\sqrt[3]{2} = 9\sqrt[3]{2}\left(\sqrt[3]{8} - 1\right) = 9\sqrt[3]{2} \neq 0$. So the tangent at $\left(3\sqrt[3]{2},\, 3\sqrt[3]{4}\right)$ is horizontal — the top of the leaf.

(b) The equation $x^3 + y^3 = 9xy$ does not change when x and y are swapped, so the curve is symmetric in the line $y = x$, and the mirror image of a horizontal tangent is a vertical one. The vertical tangent is therefore at $\left(3\sqrt[3]{4},\, 3\sqrt[3]{2}\right) \approx (4.76, 3.78)$. Directly: vertical means the denominator $y^2 - 3x$ is 0 while the numerator is not; $y^2 = 3x$ into the curve gives, by the same algebra with the roles swapped, $y^3 = 54$. (At the origin both numerator and denominator vanish — the curve crosses itself there and has two tangents, and the formula says nothing.)

(c) The point is on the curve: $\left(\frac{9}{2}\right)^3 + \left(\frac{9}{2}\right)^3 = 2 \cdot \frac{729}{8} = \frac{729}{4}$ and $9 \cdot \frac{9}{2} \cdot \frac{9}{2} = \frac{729}{4}$. The slope: $y' = \dfrac{3 \cdot \frac{9}{2} - \frac{81}{4}}{\frac{81}{4} - 3 \cdot \frac{9}{2}} = \dfrac{\frac{54}{4} - \frac{81}{4}}{\frac{81}{4} - \frac{54}{4}} = \dfrac{-27/4}{27/4} = -1$. The tangent: $y - \frac{9}{2} = -\left(x - \frac{9}{2}\right)$, i.e. $x + y = 9$. The point lies on the axis of symmetry $y = x$, so the mirror image of the tangent is the tangent itself, which forces its slope to be 1 or -1; the computation says -1: the tangent is perpendicular to the axis of symmetry.

Problem 12 *Second derivatives by implicit differentiation.*

(Not in the lecture.) To find y'': differentiate y' again, with the product or quotient rule, and replace y' by the expression already found; use the equation of the curve to simplify.

(a) $x^2 - 2y^2 = 2$: find y' and y'' in terms of x and y, simplify y'' using the equation of the curve, and evaluate both at $(2, 1)$.

(b) $y^2 = x^3$: find y' and y'' at the point $(4, 8)$.

Solution.

(a) *First derivative.* $2x - 4y\,y' = 0$, so $y' = \dfrac{2x}{4y} = \dfrac{x}{2y}$. At $(2,1)$ — on the curve, $4 - 2 = 2$ — $y' = \dfrac{2}{2} = 1$. *Second derivative.* Differentiate $y' = \dfrac{x}{2y}$ by the quotient rule, remembering that y brings a y': $y'' = \dfrac{2y \cdot 1 - x \cdot 2y'}{(2y)^2} = \dfrac{2y - 2x\,y'}{4y^2}$. Now replace y' by $\dfrac{x}{2y}$: $2x\,y' = \dfrac{x^2}{y}$, so

$$y'' = \frac{2y - \frac{x^2}{y}}{4y^2} = \frac{2y^2 - x^2}{4y^3}.$$

The curve says $x^2 - 2y^2 = 2$, i.e. $2y^2 - x^2 = -2$: $y'' = \dfrac{-2}{4y^3} = -\dfrac{1}{2y^3}$. At $(2,1)$: $y'' = -\dfrac{1}{2}$. (On the upper half of the curve, $y > 0$, we get $y'' < 0$: it bends downwards everywhere, as the graph of $y = \sqrt{x^2/2 - 1}$ does.)

(b) $2y\,y' = 3x^2$, so $y' = \dfrac{3x^2}{2y}$, and at $(4,8)$ — on the curve, $64 = 64$ — $y' = \dfrac{48}{16} = 3$. Differentiate again, quotient rule, with $(2y)' = 2y'$:

$$y'' = \frac{6x \cdot 2y - 3x^2 \cdot 2y'}{(2y)^2} = \frac{12xy - 6x^2y'}{4y^2}.$$

At $(4,8)$, with $y' = 3$: $y'' = \dfrac{12 \cdot 4 \cdot 8 - 6 \cdot 16 \cdot 3}{4 \cdot 64} = \dfrac{384 - 288}{256} = \dfrac{96}{256} = \dfrac{3}{8}$. (Replacing y' by $\frac{3x^2}{2y}$ in general and using $y^2 = x^3$ gives $y'' = \dfrac{3x^4}{4y^3}$; at $(4,8)$ that is $\frac{768}{2048} = \frac{3}{8}$ again. Here one can also solve: $y = x^{3/2}$ on the upper branch, $y'' = \frac{3}{4}x^{-1/2} = \frac{3}{8}$ at $x = 4$ — a check.)

Challenge *(for those who finish early, or at home)*

1. Problem 5(c) gave $p(x) = x|x|$ with $p'(x) = 2|x|$. Show that p is differentiable everywhere but p' is not differentiable at 0: a function can be differentiable once and not twice.
2. Let $f(x) = \dfrac{1}{1-x}$. Compute f', f'' and f''', guess a formula for $f^{(n)}(x)$, and prove it by induction.
3. Let $f(x) = x^2 \sin\dfrac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. Show from the definition that $f'(0) = 0$ (Squeeze Theorem). Then compute $f'(x)$ for $x \neq 0$ and show that f' has no limit at 0: f is differentiable everywhere, yet f' is *not continuous* at 0.

Solution.

1. Problem 5(c) showed $p'(0) = 0$ and $p'(x) = 2x$ for $x > 0$, $-2x$ for $x < 0$: so $p'(x) = 2|x|$ for every x, and p is differentiable everywhere. Is p' differentiable at 0? Its difference quotient at 0 is $\dfrac{p'(0+h) - p'(0)}{h} = \dfrac{2|h|}{h}$, so $\lim\limits_{h\to 0^+} \dfrac{2|h|}{h} = \lim\limits_{h\to 0^+} 2 = 2$ and $\lim\limits_{h\to 0^-} \dfrac{2|h|}{h} = \lim\limits_{h\to 0^-} (-2) = -2$: the one-sided limits differ, so $p''(0)$ does not exist. The graph of $p' = 2|x|$ has a corner at 0, exactly like $|x|$. So p is differentiable once everywhere, and twice everywhere except at 0. □ (Multiplying $|x|$ by x smoothed the corner once; it takes $x^2|x|$ to get two derivatives at 0, and so on.)

2. Write $f(x) = (1-x)^{-1}$ and use the chain rule with inside $1 - x$, whose derivative is -1: $f'(x) = -1 \cdot (1-x)^{-2} \cdot (-1) = (1-x)^{-2}$; $f''(x) = -2(1-x)^{-3} \cdot (-1) = 2(1-x)^{-3}$; $f'''(x) =$

$-6(1-x)^{-4}\cdot(-1) = 6(1-x)^{-4}$. The pattern: $1,\ 2,\ 6 = 3!$ on top, and the exponent one more than the order: *guess* $f^{(n)}(x) = \dfrac{n!}{(1-x)^{n+1}}$. **Proof** (by induction). It holds for $n = 1$. If it holds for some n, then $f^{(n+1)}(x) = \dfrac{d}{dx}\big[n!\,(1-x)^{-(n+1)}\big] = n!\cdot\big(-(n+1)\big)(1-x)^{-(n+2)}\cdot(-1) = (n+1)!\,(1-x)^{-(n+2)} = \dfrac{(n+1)!}{(1-x)^{n+2}}$, which is the formula with $n+1$ in place of n. So it holds for every $n \geq 1$. □

3. *At 0, from the definition:* $\dfrac{f(0+h)-f(0)}{h} = \dfrac{h^2\sin(1/h)}{h} = h\sin\dfrac{1}{h}$, and $\left|h\sin\tfrac{1}{h}\right| \leq |h|$ because $|\sin t| \leq 1$ for every t. Since $-|h| \leq h\sin\tfrac{1}{h} \leq |h|$ and $\pm|h| \to 0$, the Squeeze Theorem gives $\lim_{h\to 0} h\sin\tfrac{1}{h} = 0$: $f'(0) = 0$. *Away from* 0*:* product rule and chain rule, with $\big(\sin\tfrac{1}{x}\big)' = \cos\tfrac{1}{x}\cdot\big(-\tfrac{1}{x^2}\big)$:

$$f'(x) = 2x\sin\frac{1}{x} + x^2\cos\frac{1}{x}\cdot\left(-\frac{1}{x^2}\right) = 2x\sin\frac{1}{x} - \cos\frac{1}{x} \qquad (x \neq 0).$$

No limit at 0*:* the first term, $2x\sin\tfrac{1}{x}$, tends to 0 (squeezed by $\pm 2|x|$). At $x = \tfrac{1}{2k\pi}$ $(k = 1, 2, 3, \dots)$ we have $\sin\tfrac{1}{x} = 0$ and $\cos\tfrac{1}{x} = 1$, so $f'(x) = -1$ exactly; at $x = \tfrac{1}{(2k+1)\pi}$ we have $\sin\tfrac{1}{x} = 0$ and $\cos\tfrac{1}{x} = -1$, so $f'(x) = +1$ exactly — and both kinds of point lie as close to 0 as we like. So f' takes the values -1 and $+1$ infinitely often near 0, and $\lim_{x\to 0} f'(x)$ does not exist. In particular $\lim_{x\to 0} f'(x) \neq f'(0) = 0$: f' is not continuous at 0, although f is differentiable at every number. □ Differentiable does not imply "the derivative is continuous".